

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

Nidhin Pattaniyil · Semih Yagli · Tanwir Zaman

Independent Researchers, Hoboken, New Jersey, USA · team npatta01

11

retrieval branches

146

ranker features

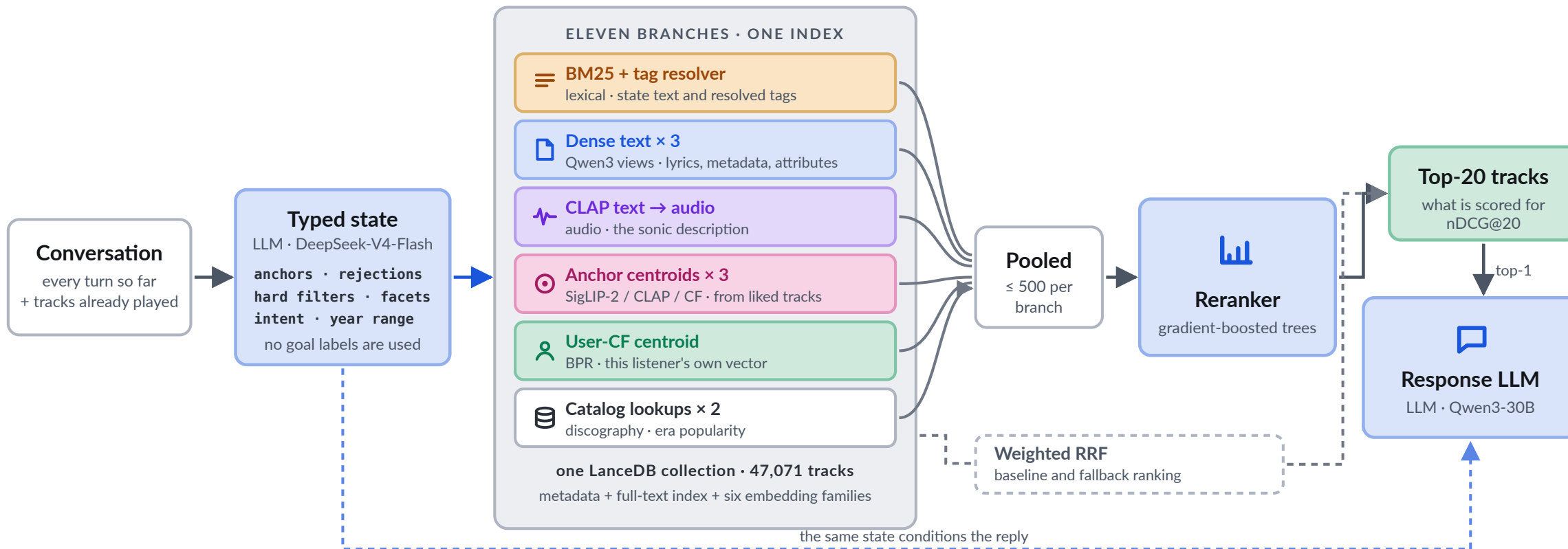
0.3811

Blind-B composite

29_{/40}

where that finished

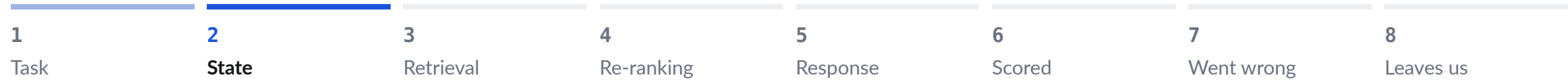
One state, eleven branches, one learned ranker



1	2	3	4	5	6	7	8
Task	State	Retrieval	Re-ranking	Response	Scored	Went wrong	Leaves us

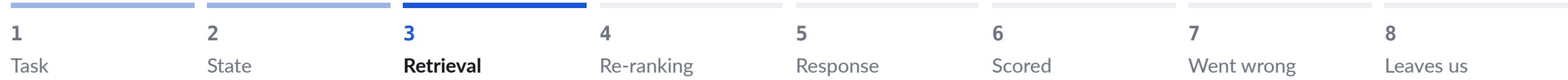
1 The task

what the challenge asks, and how it is scored



2 State extraction

reading a typed state off the conversation



3 Retrieval

eleven branches over one index



4 Re-ranking

one learned model over the pooled candidates

What the ranker leans on

FEATURE FAMILY	#	GAIN
Bi-encoder cosine	1	59.0%
Per-branch rank and score	66	11.5%
Session and artist affinity	18	10.7%
Popularity · state · tags	31	9.2%
Other similarity cosines	17	8.9%
Constraints and rejections	13	0.7%

SETUP

LightGBM LambdaMART, a gradient-boosted tree ranker. LR 0.025, 127 leaves, truncation 200, 5-fold user-grouped CV. ~20.6M candidate rows.

READ IT CAREFULLY

Split gain describes the fitted model, not recommendation quality. But **constraints at 0.7%** is real: the ranker leans on similarity, not on what the listener asked for.

GUARDS AFTER SCORING

Exact-track pin when a specific track is named; a final artist-constraint check.



5 Response generation

writing the reply — 30% of the composite

Four ways the reply went wrong

ASKED	<i>"I wanted Watercolors by Pat Metheny specifically. Can you play that one?"</i>
REPLIED	<i>"Got it – here's Alfie by Pat Metheny, a smooth, introspective piece..."</i>
JUDGE	constraint violation – claims to comply with a request it did not meet
ASKED	<i>"...something with a strong social or political message."</i>
REPLIED	<i>"Sorry, Cthulhu Dawn by Cradle of Filth leans more into dark fantasy... It doesn't match your request."</i>
JUDGE	talks its own top recommendation down without justifying why it was picked
ASKED	<i>"Is it the rhythm or the overall instrumentation that makes it feel so good?"</i>
REPLIED	<i>"You're really into that smooth, groove-driven nu-jazz vibe – this one's got..."</i>
JUDGE	recommends instead of answering; breaks conversational flow
ASKED	<i>"Which song from Grace/Wastelands was one of the best tracks of 2009?"</i>
REPLIED	<i>"...hauntingly beautiful and rich with folk-inflected melancholy... raw emotion and experimental edge."</i>
JUDGE	generic praise, no evidence from the conversation or the track

The middle two are the prompt **working as written** — it was told to flag a conflict rather than oversell. We tuned it against the Blind-A judge and never asked what it did to the recommendation it was describing.



6 What it scored

the official splits, and what we steered on

How it scored

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A • development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B • final, 29th of 40	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

category and lexical barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



7 What went wrong

four findings, and the turns behind them

What is not ideal

We tuned on what we measured.

The 8,000-turn development set was both our training and our evaluation data, and the response prompt was picked against the Blind-A leaderboard: 4.70 there, 3.30 on Blind-B.

We extracted more than we used.

The state we read off a conversation was far richer than the state that reached retrieval or ranking. Most fields never became a filter or a ranking signal at all.

Nothing from outside the catalog.

No outside source for what a track or artist is known for, and no tempo, key or soundtrack. Past one CF vector, no listening behaviour — no co-occurrence, no transitions.

The reply was never checked.

Asked for a track we do not have, the generator described a different song by the same artist instead of saying so.

Asked for, served

the Kamelot track
from **Silverthorn**



Kamelot – Fallen Star, from **Haven**
CONSTRAINT NEVER APPLIED

exactly **126.70 bpm**,
in G minor



no tempo or key field exists to match on
NOTHING IN THE CATALOG

“Watercolors”
by Pat Metheny



Metheny – Alfie, described as if it were
RESPONSE LLM

Silverthorn is in the catalog. The third is a response failure, not a retrieval one – the track simply does not exist in the 47,071.

Failure 1 — the constraint was extracted, then ignored

ANALYSIS CORE

USER *"...I gotta break away from Deltron and Del completely. Looking for **new** established underground hip-hop artists."*

STATE avoid **Deltron 3030**, **Del The Funky Homosapien** · request_type **new_artist**

WE SERVED **Dead Prez – Hip Hop** (classic hip-hop, the thing they rejected)

USER *"...branching out to other singer-songwriter or alt-country artists."*

STATE avoid **Ryan Adams & The Cardinals** · request_type **new_artist**

WE SERVED **Ryan Adams – When The Stars Go Blue**

USER *"...still focusing on Juanes. While I enjoy his music, I haven't found the specific lyrical depth I'm after."*

STATE rejected **Juanes** · avoid **Juanes**, **Latin Pop**

WE SERVED **Juanes – Me Enamorado De Ti**

Three different sessions, one failure: the listener asked to move on, the state recorded it, and **the ranker served that artist anyway**. Constraint features carry 0.7% of split gain — nothing in the model was weighing them.

What the training labels look like

WENT WRONG **CORE**

THE LISTENER ASKED	JUST PLAYED	GROUND-TRUTH NEXT TRACK	ANNOTATED
"I keep asking for <i>My Body Is a Cage</i> by Arcade Fire ... that's the only song I want"	Arctic Monkeys	Arctic Monkeys – This House is a Circus	DOES NOT
"...something with a similar chill vibe, but from a different artist"	Bonobo	Bonobo – Jets	MOVES
"What other artists or tracks fit this introspective, darkwave style?"	Tropic of Cancer	Tropic of Cancer – The Dull Age	MOVES
"What other powerful orchestral tracks or artists would you recommend next?"	Nick Phoenix	Nick Phoenix – Kogan	MOVES

Three of these four are annotated **MOVES TOWARD GOAL** — the dataset records them as the recommendation having helped. Train on this and the model learns to repeat the just-played artist. **Ours does.**

The target often ignores the request

User: "...something with a similar chill, electronic vibe, but **from a different artist?**" (just played: Bonobo)

Target: **Bonobo – "Jets"** · annotated MOVES toward goal

Next turn: "'Jets' is cool, but I was actually hoping for something **from a different artist this time...**"

Target: **Bonobo – "Pieces"** · annotated MOVES toward goal

Ground truth is simply **whatever played next**, in a session an LLM wrote. Our own model anchors on the just-played artist — we think it learns that here.



8 Where it leaves us

what we would change, and what is released

What we would do differently

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.

Everything is released

IN THE REPO

- Pipeline code, configs, trained models
- Cached per-turn extraction states
- The 113,393 re-judged labels
- A frozen-LLM-cache replay bundle

ZERO-CREDENTIAL REPLAY

Every LLM call is served from a curated frozen cache that **hard-fails on a miss** rather than falling through to a paid call. Both submissions reproduce without any API key.



Code, models, data

[github.com/npatta01/
music-crs-2026](https://github.com/npatta01/music-crs-2026)

